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MACHINE LEARNING FOR DATA SCIENCE

Customer Attrition Analysis and
Prediction in the Banking Sector using
Machine Learning Classifiers

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1. Introduction

Customer attrition, also known as customer churn, is an important issue in the banking sector because losing existing customers can reduce long-term profitability and increase the cost of customer acquisition (Hon et al., 2023; Singh et al., 2024). Compared with attracting new customers, retaining existing customers is often more cost-effective for financial institutions. Therefore, identifying customers who are likely to leave the bank can help managers design proactive retention strategies and improve customer relationship management.

Customer churn is usually influenced by multiple factors, such as demographic characteristics, credit score, account balance, product usage, tenure, and customer activity status. These factors may not affect churn independently, as the relationship between customer attributes and churn behavior can be complex and non-linear. For this reason, machine learning techniques are useful for analyzing historical customer data and identifying churn patterns that may not be easily observed through manual analysis (Tran et al., 2023).

This project focuses on customer attrition analysis and prediction in the banking sector using supervised machine learning classifiers. The dataset used in this study is obtained from Kaggle and contains 10,000 customer records with 14 attributes, including credit score, geography, gender, age, tenure, balance, number of products, activity status, estimated salary, and churn status (Mathchi, n.d.). The target variable is “Exited”, which indicates whether a customer has left the bank.

The main aim of this project is to develop a predictive model that can identify customers at high risk of churn. To achieve this aim, exploratory data analysis is first conducted to understand the distribution of customer characteristics and their relationship with churn. Then, three supervised machine learning algorithms, namely Logistic Regression, Random Forest, and Support Vector Machine, are implemented and compared. The performance of these models is evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC, with particular attention given to the identification of churn customers as the minority class.

2. Background and Literature Review

Churn prediction is an important task in banking because losing existing customers reduces long-term profitability and increases the cost of attracting new customers (Hon et al., 2023; Singh et al., 2024). In banking, churn usually refers to a customer stopping using a bank's services or terminating a relationship with a bank (Hon et al., 2023). Because customer behavior can be influenced by factors such as age, credit score, account balance, tenure, product usage, and activity status, machine learning can be used to identify patterns that are difficult to observe manually.

Recent studies have shown that ensemble learning models often outperform traditional linear classifiers in churn prediction tasks because they can better capture complex customer behaviour patterns and feature interactions (Boozary et al., 2024; Ashraf, 2024).

Previous studies have generally treated bank churn prediction as a supervised classification problem (Tran et al., 2023; Hon et al., 2023; Singh et al., 2024). Logistic Regression is often used as a baseline model because it is simple, interpretable, and suitable for binary classification. However, it may not perform well when customer behavior exhibits complex or non-linear patterns. Random Forest is often applied because it can handle feature interactions and reduce weaknesses in individual decision trees. Support Vector Machine is also used for binary classification because it can separate churn and non-churn customers through an optimal decision boundary. Therefore, these models are suitable for comparison in this project. Feature engineering has also been widely recognised as an important step in improving predictive performance. By creating meaningful derived variables, machine learning models can better capture customer behaviour patterns and interactions among financial attributes (Abedin et al., 2023; Zheng & Casari, 2018).

Class imbalance is a common challenge in churn prediction because churn customers typically represent a minority proportion of the dataset. Previous studies have shown that resampling techniques such as SMOTE can improve the detection of minority-class customers and enhance Recall and F1-score performance (Chawla et al., 2002; Ashraf, 2024).

A key issue in churn prediction is class imbalance, where non-churn customers often outnumber churn customers (Hon et al., 2023). Because of this, accuracy alone may not fully reflect model performance. A model may achieve high accuracy by correctly predicting most non-churn customers, but it may still fail to detect customers who actually churn. Therefore, this project evaluates the models using Accuracy, Precision, Recall, F1-score, and ROC-AUC, with a particular focus on the models' ability to identify churn customers. This evaluation approach is consistent with the project objectives, which focus on comparing Logistic Regression, Random Forest, and SVM using Accuracy, F1-score, and ROC-AUC. Recent studies have also highlighted the importance of feature engineering and class balancing techniques in improving churn prediction performance, particularly when the target class is imbalanced (Abedin et al., 2023; Chawla et al., 2002).

Table 2.1 Summary of Related Studies on Customer Churn Prediction

Study	Methods Used	Main Focus	Relevance to This Project
Tran et al. (2023)	Logistic Regression, Decision Tree, Random Forest, SVM, KNN	Compared multiple machine learning models for bank customer churn prediction.	Supports the use of several classifiers instead of relying on one model.
Hon et al. (2023)	KNN, SVM, Naïve Bayes, Decision Tree, Random Forest, XGBoost	Evaluated multiple metrics, including Accuracy, F1-score, and ROC-AUC.	Supports the evaluation metrics used in this project.
Singh et al. (2024)	Logistic Regression, SVM, Random Forest, XGBoost	Applied machine learning to identify customers is likely to discontinue banking services.	Shows that churn prediction is a practical classification problem in banking.
Boozary et al. (2024)	Logistic Regression, SVC, Decision Tree, XGBoost, Gradient Boosting	Compared different classifiers for bank churn prediction.	Supports model comparison and the use of both simple and advanced classifiers.

3. Methodology

3.1 Dataset Description

This project used a secondary dataset obtained from Kaggle, titled *Churn for Bank Customers* (Mathchi, n.d.). The dataset contains 10,000 customer records and 14 original attributes. The target variable is Exited, where a value of 1 means the customer has left the bank and a value of 0 means the customer remains with the bank. The dataset includes customer-related features such as CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. Before model development, identifier columns, including RowNumber, CustomerId, and Surname, were removed because they did not provide meaningful predictive information. After these columns were removed, the remaining features were used for exploratory data analysis and model training.

3.2 Data Preprocessing

The dataset was first examined to understand its structure, data types, missing values, and feature characteristics. No missing values were found in the selected variables, so imputation was not required. Categorical variables, including geographic location and gender, were later encoded as part of the feature engineering process described in Section 3.4.

Feature engineering was conducted after EDA. The dataset was then divided into training and test sets using an 80:20 ratio with stratified sampling. StandardScaler was subsequently applied to the training set and used to transform the test set, ensuring no data leakage from the test set.

3.3 Exploratory Data Analysis

In order to understand the distribution of customer characteristics and their relationship with customer churn, exploratory data analysis was conducted. The analysis focused on numerical and categorical variables. The target variable distribution was first examined to identify class imbalance between churn and non-churn customers. Numerical features such as Age, CreditScore, Tenure, Balance, NumOfProducts, and EstimatedSalary were analysed using distribution plots and boxplots. Churn rate plots were used to compare categorical features, including Geography, Gender, HasCrCard, and IsActiveMember. Correlation analysis was also conducted to examine the relationship between customer attributes and churn.

3.4 Feature Engineering & Target Class Balancing

Feature engineering was conducted after exploratory data analysis to create additional variables that could better represent customer financial behaviour and engagement. Three engineered features were initially created: BalancePerProduct, ActiveBalance, and AgeGroup.

BalancePerProduct was calculated by dividing customer balance by the number of products owned plus one. This feature was designed to represent the average balance associated with each banking product.

ActiveBalance was created by multiplying Balance with IsActiveMember. This feature combines financial value with customer activity status and helps capture differences between active and inactive customers with similar balances.

Age was later removed before model training because we already have the AgeGroup. Class balancing was done using SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features) over SMOTE as our data contains a mixture of continuous and categorical features.

3.5 Machine Learning Models

In this project, three supervised machine learning classifiers were developed and compared: Logistic Regression, Random Forest, and Support Vector Machine.

Logistic Regression was used as a baseline model because it is suitable for binary classification and provides an interpretable comparison point. Random Forest was selected as an ensemble learning model because it can capture non-linear relationships and interactions between variables. SVM was included because it can construct a decision boundary that separates churn and non-churn customers.

The best model will undergo a round of hyperparameter tuning and cross-validation to further improve the model metrics.

3.6 Model Evaluation

The models were evaluated using the test set. Several performance metrics were used, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy measures the overall proportion of correct predictions. However, accuracy alone is not sufficient to evaluate model performance due to an imbalanced dataset. Precision measures the proportion of customers predicted as churn who actually churned, while Recall measures the proportion of actual churn customers correctly identified by the model. F1-score provides a balance between Precision and Recall, making them suitable for evaluating minority class predictions. ROC-AUC is used to evaluate the model's ability to distinguish between churned and non-churned customers at different classification thresholds.

Since the main objective of this project was to identify customers at high risk of churn, greater attention was given to Recall and F1-score for the churn class. A model with high Accuracy but low

Recall may fail to identify many actual churn customers, which reduces its practical usefulness for customer retention.

3.7 Feature Importance Analysis

To improve model interpretability, feature importance analysis was conducted using the Random Forest model. Random Forest was selected for interpretation because it can estimate the relative contribution of each feature to the prediction process. This analysis was used to identify the most influential variables affecting customer churn prediction.

4 Results and Discussion

4.1 Exploratory Data Analysis Results

The target variable distribution shows that the dataset was imbalanced. Among the 10,000 customers, approximately 79.63% were retained, while 20.37% churned. This indicates that churn customers formed the minority class. Therefore, model evaluation should not rely only on Accuracy, as a model may achieve high Accuracy by mainly predicting the majority non-churn class.

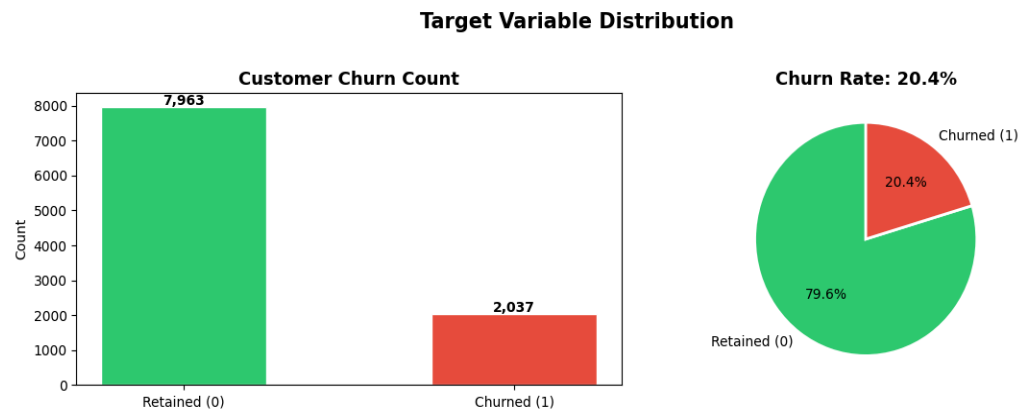


Figure 4. 1 Target Variable Distribution

Correlation analysis showed that Age had the strongest positive correlation with customer churn, followed by Balance. This suggests that older customers and those with higher account balances are more likely to leave the bank. In contrast, IsActiveMember was negatively correlated with churn, suggesting that active members were less likely to leave. These findings suggest that both demographic and account activity characteristics correlate with predicting customer churn.

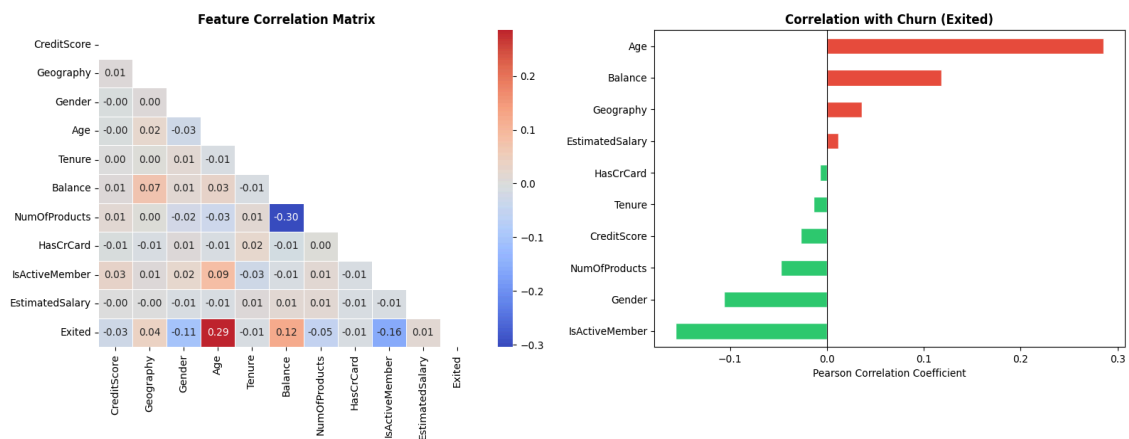


Figure 4. 2 Correlation Analysis of Customer Attributes and Churn

Categorical feature analysis further supports the importance of customer background and activity status. Customers from different geographic regions exhibited different churn rates, while inactive members were more likely to churn. These patterns suggest that churn is not random but is associated with identifiable customer characteristics.

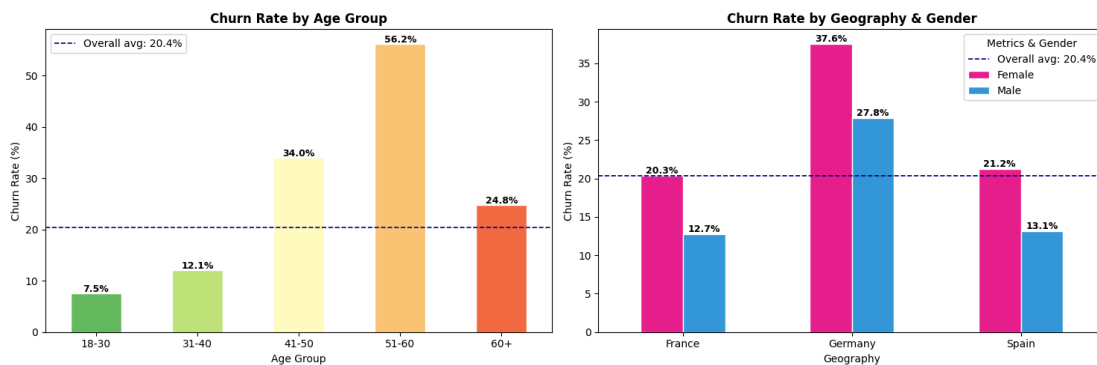


Figure 4. 3 Churn Rate by Age Group and Geography-Gender Analysis

Further analysis using age groups revealed a strong relationship between customer age and churn behaviour. Customers aged 51–60 recorded the highest churn rate (56.2%), followed by customers aged 41–50 (34.0%). In contrast, younger customers aged 18–30 exhibited the lowest churn rate (7.5%). These findings further support the importance of age-related features in churn prediction and motivated the creation of the AgeGroup feature during feature engineering.

4.2 Class Balancing Results

Table 4.1 Class Imbalance Handling (SMOTENC)

Dataset	Non-Churn	Churn
Before SMOTENC	6370	1630
After SMOTENC	6370	6370

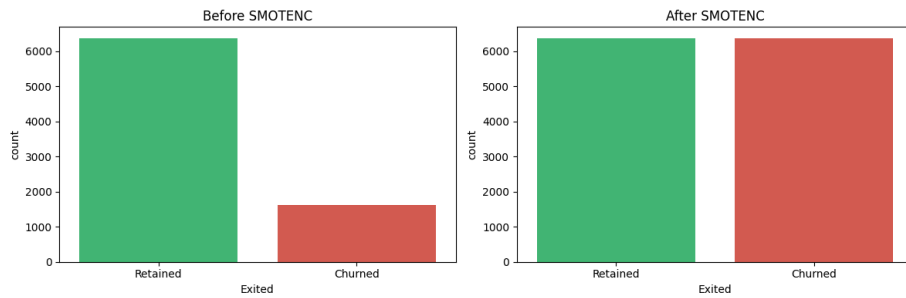


Figure 4. 4 Class Distribution Before and After SMOTENC

SMOTENC successfully balanced the minority churn class by increasing the number of churn samples from 1,630 to 6,370, matching the number of retained customers. This helped reduce model bias toward the majority class and improved the ability of the classifiers to identify churn customers. Balancing the training data was particularly important because Recall and F1-score were key evaluation metrics in this project.

4.3 Model Performance Comparison

Table 4.2 Performance Comparison of Machine Learning Classifiers

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.8285	0.5711	0.6315	0.5998	0.8439
SVM	0.7910	0.4908	0.7224	0.5845	0.8406
Logistic Regression	0.7175	0.3903	0.6904	0.4987	0.7731

Random Forest achieved the highest Accuracy, F1-score, and ROC-AUC, indicating the strongest overall predictive performance. Its F1-score of 0.5998 shows that it provided the best balance between Precision and Recall among the three models.

SVM achieved the highest Recall of 0.7224, meaning it was the most effective model in identifying actual churn customers. However, its Precision and F1-score were lower than Random Forest.

Logistic Regression achieved the lowest overall performance, although its Recall was still acceptable after applying SMOTENC. This indicates that SMOTENC helped the baseline model identify more churn customers, but Logistic Regression was less effective than Random Forest and SVM in overall classification performance.

4.4 Hyperparameter Optimization

Random Forest was selected for hyperparameter tuning to further improve the model performance. A Bayesian search was conducted across key hyperparameters. The primary optimization metric selected was the cross-validated F1-score. Selection of parameters as per table below.

Table 4.3 Hyperparameter Search Space and Optimal Pipeline Configuration

Hyperparameter	Search Range	Best Value	Justification
smotenc__k_neighbors	3-10	4	Determines the neighborhood size for generating localized synthetic minority samples.
rf__n_estimators	100-600	100	Controls the number of ensemble decision trees to stabilize variance.
rf__max_depth	3-30	7	Limits tree depth to control underfitting versus majority-class overfitting.

rf_min_samples_split	2-20	20	Minimum samples required to split an internal node and act as a regularizer.
rf_min_samples_leaf	1-15	1	Minimum samples required at a leaf node to control minority instance isolation
rf_max_features	“sqrt”, “log2”, None	“sqrt”	Features evaluated at each split to inject randomness to de-correlate trees.
rf_class_weight	“balanced”, “balanced_subsample”, None	“balanced”	Applies cost-sensitive penalties inversely proportional to class frequencies.
rf_bootstrap	True, False	False	Determines whether trees are trained on bootstrapped or complete samples.

The optimized pipeline achieved a best cross-validated F1-score of 0.5923. When tested against the test set, the fine-tuned model performs better in Recall and ROC-AUC compared to the base model while Accuracy and Precision did show a slight decline. F1-Score remains almost similar between the two. The table below shows the comparison of the metrics.

Table 4.4 Performance Comparison of Based and Tuned Random Forest Classifier

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest (Fine-tuned)	0.7975	0.5017	0.7273	0.5938	0.8547
Random Forest	0.8285	0.5711	0.6315	0.5998	0.8439

4.5 Confusion Matrix and ROC Curve Analysis

The confusion matrices provide a clearer comparison of how each model classified churn and non-churn customers.

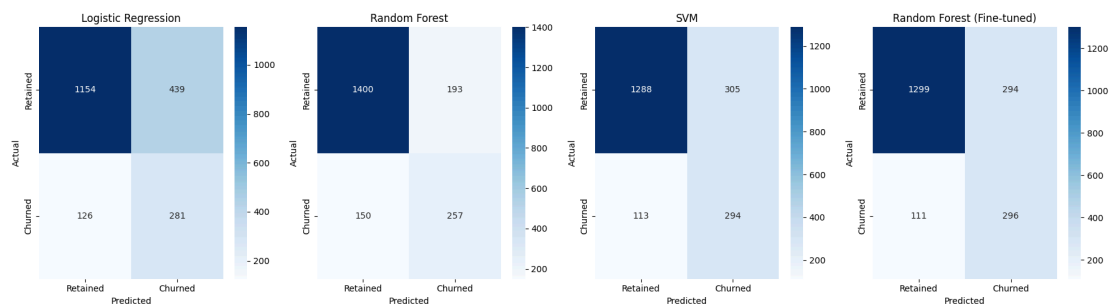


Figure 4.5 Confusion Matrices of Four Models

Random Forest correctly classified the most retained customers, achieving the highest overall Accuracy. However, compared to SVM, its Recall for the churn class was lower (0.6315 vs. 0.7224), indicating that it missed more actual churn customers. This trade-off is reflected in its higher precision, suggesting that when Random Forest predicts a customer will churn, it is more likely to be correct.

SVM showed a better balance between detecting churn customers and maintaining acceptable overall classification performance. Although it was less accurate than Random Forest, it identified more actual churn customers. This indicates that SVM was more effective in identifying churn customers. However, Random Forest maintained stronger overall classification performance across multiple evaluation metrics.

Ultimately, the fine-tuned Random Forest which was optimized via the SMOTENC and class-weighting pipeline delivers the most performance. By effectively penalizing majority-class bias and learning from localized synthetic minority clusters, it successfully minimizes False Negatives to the lowest across all models (111) while maximizing True Positives (296). Although this comes at the expense of a slight increase in False Positives (294) compared to its baseline counterpart, this is a highly acceptable, deliberate trade-off that fulfills the primary objective of mitigating the severe class imbalance and capturing the maximum number of at-risk customers.

The ROC curve comparison below shows that fine-tuned Random Forest achieved the highest ROC-AUC value of 0.8547, followed closely by base Random Forest with 0.8439. Logistic Regression obtained the lowest ROC-AUC value of 0.7618. These results indicate that both fine-tuned and base Random Forest had the strongest overall ability to distinguish between churned and retained customers, while SVM also showed competitive discriminatory performance.

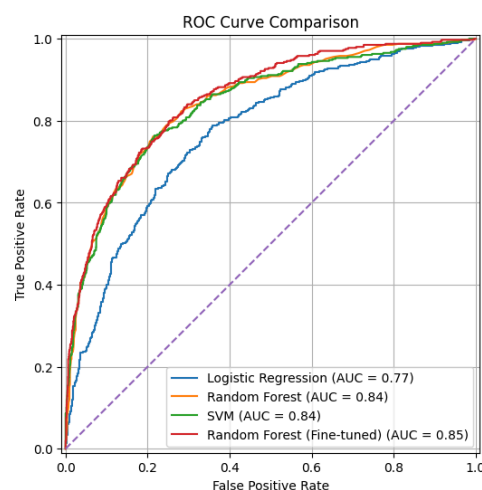


Figure 4. 6 ROC Curve Comparison

Ultimately, this confirms that the integration of SMOTENC and class-weighting did not merely forcefully shift the decision boundary to capture more True Positives, but fundamentally improved the model's underlying ability to distinguish the minority churn patterns from the majority class noise.

4.6 Feature Importance Analysis

The Random Forest feature importance analysis showed that **AgeGroup** was the most influential predictor of customer churn, followed by **NumOfProducts**, **Geography_Germany**, **Balance**, and **BalancePerProduct**. This indicates that customer demographics, product usage, financial profile, and customer relationship duration were important in predicting churn.

In addition to the top-ranked predictors, **EstimatedSalary**, **Gender_Male**, and **IsActiveMember** also appeared among the ten most important features. These results indicate that customer financial characteristics, geographical location, and engagement level all contribute to churn behaviour.

The inclusion of **ActiveBalance** and **BalancePerProduct** among the top ten features suggests that feature engineering successfully generated useful information beyond the original dataset variables. These engineered variables helped capture customer engagement and product utilisation behaviour that may not have been directly observable from the raw features.

Therefore, the feature importance results support the value of both original features and engineered features in improving churn prediction. The dominance of **AgeGroup** in the feature importance ranking is consistent with the earlier EDA findings, where older customer groups exhibited substantially higher churn rates. This consistency strengthens confidence in the model interpretation and suggests that age-related characteristics play a critical role in customer attrition.

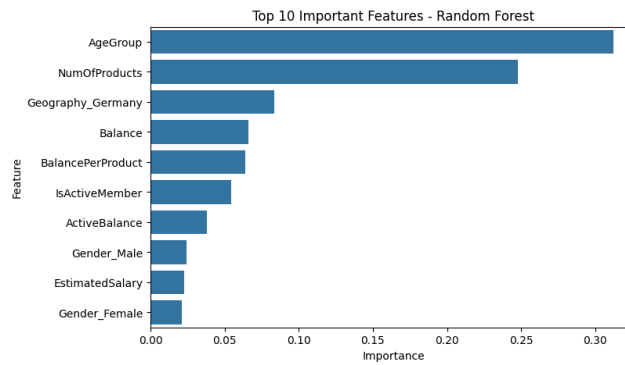


Figure 4. 7 Random Forest Feature Importance

4.7 Final Model Selection

Based on the evaluation results, the Fine-tuned Random Forest was selected as the final predictive model for this project. It achieved the highest Recall (0.7273) and the highest overall class separability with an ROC-AUC of 0.8547. While the baseline Random Forest yielded a marginally higher overall F1-score (0.5998 vs. 0.5938) and Accuracy (0.8285 vs. 0.7975), it achieved this by favoring the majority class, resulting in a critically lower Recall (0.6315). The fine-tuned model's superior Recall and ROC-AUC indicate that it successfully fulfills the primary business objective which is maximizing the identification of at-risk churners while maintaining reliable probability classification.

Although the SVM model also demonstrated a strong ability to detect churners with a highly competitive Recall of 0.7224, the Fine-tuned Random Forest consistently outperformed it across all key metrics. It yielded better Precision (0.5017 vs. 0.4908), a higher F1-score (0.5938 vs. 0.5845), and better overall separability (ROC-AUC of 0.8547 vs. 0.8406).

Therefore, the Fine-tuned Random Forest was selected as the optimal final model, providing the most robust and stable performance for capturing potential churners in a severely imbalanced context.

5 Conclusion

The results showed that Random Forest achieved the highest Accuracy, F1-score, and ROC-AUC, making it the best overall predictive model in this study. SVM achieved the highest Recall, meaning it was more effective in detecting actual churn customers, but Random Forest provided better overall balance between Precision and Recall.

The use of SMOTENC helped address the class imbalance problem by balancing the training data, while feature engineering improved the representation of customer financial behaviour and engagement. The feature importance results showed that Age, NumOfProducts, EstimatedSalary, CreditScore, Tenure, BalancePerProduct, and ActiveBalance were relevant factors in churn prediction.

Overall, this project demonstrated that combining feature engineering, SMOTENC, machine learning classifiers, hyperparameter tuning and feature importance analysis can effectively support customer churn prediction in the banking sector. From a business perspective, the findings can help banks identify high-risk customers earlier and implement targeted retention strategies, thereby reducing customer attrition and improving long-term profitability.

This study has several limitations. First, the analysis was conducted using a single publicly available dataset, which may not fully represent customer behaviour across different banking institutions. Second, only three machine learning classifiers were evaluated. Future research may explore advanced ensemble methods such as XGBoost, LightGBM, and CatBoost, together with hyperparameter optimization techniques to further improve predictive performance.

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